Pyramid of numbers

Charles Elton used a pyramid as a way of graphically representing the different levels in a food chain, with the producers at the bottom, the primary consumers on the level above, and so on. Often, the primary consumers—insects, in particular—will outnumber the producers, but the higher levels of consumers will become less numerous toward the top of the pyramid. This system does not take account of parasites; fleas and ticks on mammals and birds will far outnumber the total of all the vertebrates in an ecosystem.

In 1938, German-born animal ecologist Frederick Bodenheimer modified Elton's pyramid of numbers to produce a pyramid of biomass that represented the amount of living matter in a given area at every level. This took into account the fact that some organisms are much larger than others, but because it showed comparative biomasses at a fixed point in time, it produced anomalies. For example, in a pond, the mass of the phytoplankton producer (microscopic organisms that are the foundation of the aquatic food web) may not be as great as the mass of the fish consumers at a particular point in time, so the pyramid will be inverted. However, phytoplankton reproduce quickly when conditions, such as sunlight and nutrients, are right. Over time, the mass of the phytoplankton will far outweigh that of the fish.